

Spotify Track Popularity Analysis

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1 Introduction

In today’s music industry, achieving popularity on platforms like Spotify can make or break an artist’s career. But what makes a song go viral? Is it the energy of the track, the positivity of the sound, or the tempo that keeps listeners hooked?

This project explores the relationship between the audio characteristics of songs and their popularity on Spotify. Using a combination of an open-source Kaggle dataset and real-time Spotify API data, the aim is to answer a central research question: **What audio features are most predictive of a song’s popularity?**

Understanding these relationships can provide valuable insights for artists, producers, and marketers aiming to optimize music for broader audience reach. It also sheds light on broader patterns in listener behavior — what kinds of sounds dominate global music trends, and why.

By analyzing features such as danceability, energy, valence, tempo, and others, this project seeks to uncover which aspects of a track’s audio profile are most strongly associated with listener appeal and commercial success.

2 Methods

2.1 Data Sources

Two datasets were combined:

- **Kaggle Dataset:** An open-source dataset containing track metadata and audio features (e.g., danceability, energy, valence, tempo, loudness, etc.).
- **Spotify API:** Used to fetch up-to-date popularity scores for a random sample of tracks.

The Kaggle dataset provided static metadata and feature measurements, while the API allowed access to current popularity scores for analysis.

2.2 Data Preprocessing

The data were cleaned and prepared as follows:

- **Filtering:** Tracks with a **popularity** score of 0 were removed (32% of the sample). These tracks likely represent newly released or highly obscure songs with no measurable play count and could bias the analysis if retained.
- **Feature Engineering:** Track duration was converted from milliseconds to minutes. Genre was converted to factor type to help with modeling.
- **Cleaning:** Duplicate tracks were removed, and tracks longer than 15 minutes (likely podcasts) were excluded to focus the analysis on typical music tracks.

The final dataset consisted of tracks with nonzero popularity scores and typical audio feature measurements.

2.3 Variables Description

Key audio features analyzed:

- **Genre:** Genre of the song (Hip-hop, electronic, jazz etc).
- **Danceability:** How suitable a track is for dancing (0–1 scale).
- **Energy:** Intensity and activity level of a track.
- **Valence:** Positivity conveyed by a track (0 = sad, 1 = happy).

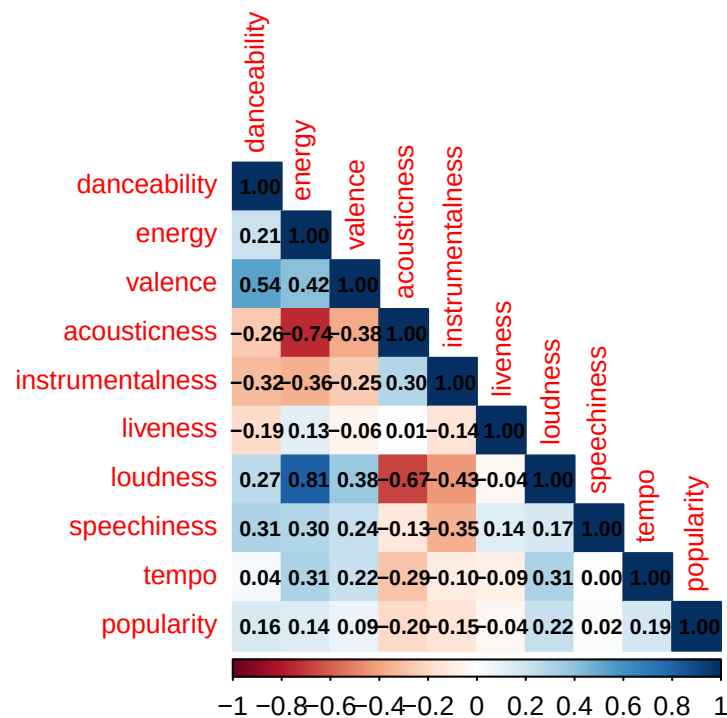
- **Tempo:** Speed or pace of a track (beats per minute).
- **Loudness:** Average decibel level across the track.
- **Speechiness:** Presence of spoken words in a track.
- **Acousticness:** Likelihood that the track is acoustic.
- **Instrumentalness:** Likelihood that a track has no vocals.
- **Liveness:** Likelihood that the track was recorded live.

The artist's name and duration of the song have been excluded from the variables of interest, because the column 'artist_name' has too many different values to gain any valuable insights (high cardinality might lead to overfitting) and the duration of a song might not be very relevant in assessing its popularity.

3 Spearman Correlation Heatmap

Since some audio features may not be normally distributed and the relationships could be monotonic but nonlinear, a Spearman correlation heatmap was constructed to better explore the strength and direction of associations between variables.

Spearman Correlation Heatmap of Audio Features



The heatmap highlights notable relationships among features. For example, **danceability** and **valence** show a moderate positive correlation, while **acousticness** and **loudness** exhibit a negative association. Popularity is modestly correlated with features like **speechiness** and **tempo**, but overall correlations remain relatively weak, reinforcing the complexity of predicting track success based solely on audio characteristics.

4 Train-Test Split for Model Evaluation

4.1 For Linear Regression

The dataset was first split into a training set and a test set for fitting and evaluating the linear regression model. No separate validation set was needed as linear regression does not involve hyperparameter tuning. The dataset was split into: - 70% training data - 30% test data.

4.2 For Tree-Based Models

For tree-based models, a three-way split was used: training set for fitting, validation set for tuning, and test set for final evaluation. This ensures that the final evaluation is unbiased by the tuning process. The data was split into: - 60% training data - 20% validation data - 20% test data.

5 Linear Regression Model

A multiple linear regression model was fitted using the training set and evaluated on the test set to assess its predictive performance.

5.0.1 Why Linear Regression?

Linear regression was chosen as a baseline model due to its interpretability and simplicity. It tests for linear associations between audio features (predictors) and track popularity (response).

5.0.2 Assumptions:

- Linearity between predictors and the outcome
- Independence of errors
- Homoscedasticity (constant variance of errors)
- Normal distribution of residuals

5.0.3 Limitations:

The true relationship between audio features and popularity may not be linear. If interactions or nonlinear effects exist, linear regression will miss them, leading to underfitting.

5.0.4 Interpretation:

The linear regression model estimated the relationship between audio features and Spotify track popularity. While several predictors had weak or statistically insignificant effects, a few variables stood out:

- **Speechiness** had a **large negative effect** on popularity ($estimate = -21.9$, $p = 0.085$), suggesting that tracks with more spoken-word content tend to be **less popular**, though the result was only marginally significant.

- **Tempo** had a **positive coefficient** ($estimate = 0.069$, $p = 0.125$), suggesting a weak upward trend: faster songs may be slightly more popular, but the effect is not strong enough to confirm statistically.
- Among the genre dummy variables, several showed **strong and statistically significant negative effects**, relative to the **baseline genre** (the first alphabetically omitted genre during dummy encoding — likely "Alternative"):
 - `genre_Children's Music` ($estimate = -31.94$, $p = 0.002$)
 - `genre_Classical` ($estimate = -21.12$, $p = 0.016$)
 - `genre_Opera` ($estimate = -30.19$, $p = 0.005$)
 - `genre_Reggaeton` ($estimate = -24.41$, $p = 0.002$)
 - `genre_Ska` ($estimate = -25.76$, $p = 0.002$)
 - `genre_Soundtrack` ($estimate = -20.98$, $p = 0.046$)
 - `genre_World` ($estimate = -24.73$, $p = 0.003$)

These results suggest that tracks in these genres tend to have **substantially lower popularity** compared to the baseline genre.

- A few other genre effects, such as `genre_Hip-Hop` ($estimate = +9.07$, $p = 0.237$), showed a **positive relationship**, though they were **not statistically significant**.

Overall, the model highlights that **genre is a strong driver of popularity**, with several genres consistently associated with lower popularity scores. Most audio features such as **danceability**, **energy**, and **valence** showed **weak or inconsistent relationships**, suggesting that **non-audio factors like genre may play a bigger role** in determining track popularity.

6 Tree-Based Machine Learning Models

6.1 Regression Tree Analysis

A regression tree model was fitted to capture potential nonlinearities and interaction effects among the predictors.

6.1.1 Why Regression Trees?

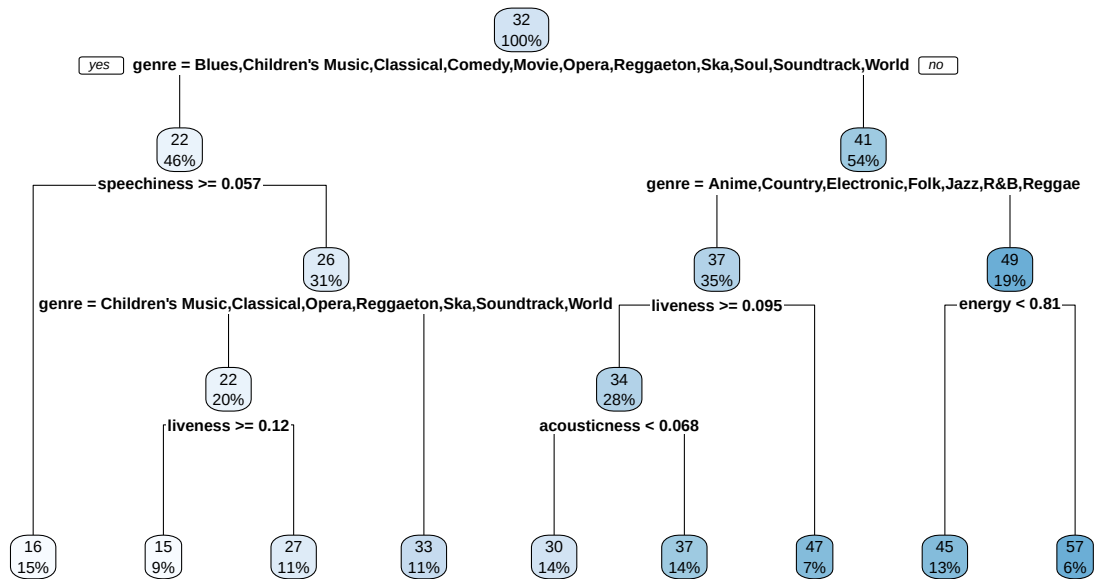
Regression trees can automatically detect and model nonlinear interactions without requiring explicit specification. They are highly interpretable, showing clear decision rules.

6.1.2 Assumptions:

- No assumptions of linearity or normality
- Data is partitioned recursively based on feature thresholds

6.1.3 Limitations:

Regression trees are prone to overfitting and can be sensitive to small changes in the data. Pruning or ensemble methods are needed to stabilize performance.



6.1.4 Interpretation:

The regression tree model reveals how combinations of **genre** and specific audio features help explain variation in track popularity. The tree splits the data into distinct segments based on conditions that best reduce the prediction error at each step.

6.1.4.1 Key takeaways:

- **Genre is the dominant predictor** in the first split. Tracks belonging to genres such as *Children's Music*, *Classical*, *Opera*, *Reggaeton*, *Ska*, *Soundtrack*, and *World* are grouped together and associated with **lower average popularity** compared to genres like *Electronic*, *Jazz*, *R&B*, and *Country*. This reinforces the findings from the linear regression model.
- Within the low-popularity genres group:
 - **Speechiness** becomes a key splitter. Tracks with high spoken content ($\text{speechiness} \geq 0.057$) are further divided by genre again.
 - Tracks from *Children's Music*, *Classical*, *Opera*, etc. with **high speechiness and low liveness** tend to have the **lowest popularity scores**, consistent with the idea that highly spoken, live-sounding music in niche genres is less appealing to mainstream audiences.
- For the remaining genres:
 - **Liveness** and **acousticness** help distinguish subgroups. Tracks with lower liveness and lower acousticness appear to cluster toward higher popularity.
 - In particular, tracks with **high energy** (≥ 0.81) are placed into the segment with the **highest predicted popularity** at the far right of the tree.

6.1.4.2 Overall insight: This tree structure provides an interpretable, rule-based understanding of track popularity. It reinforces that **genre alone is a strong discriminator**, and that certain features like **speechiness**, **liveness**, and **energy** further stratify popularity within genre-based groups.

While more flexible models like random forests outperform the tree in RMSE, the regression tree offers valuable insight into **how** and **why** different subgroups of music tracks differ in popularity.

6.2 Random Forest Analysis with Hyperparameter Tuning

A Random Forest model was trained and tuned to improve predictive performance. The best number of randomly selected predictors (`mtry`) was selected using 5-fold cross-validation.

6.2.1 Why Random Forest?

Random Forests aggregate the predictions of many trees, reducing overfitting and capturing complex non-linear interactions. They tend to perform better than individual trees on unseen data.

6.2.2 Assumptions:

- Data points are independent
- No strict assumptions on distribution or linearity
- Some randomness improves generalization (bagging and random feature selection)

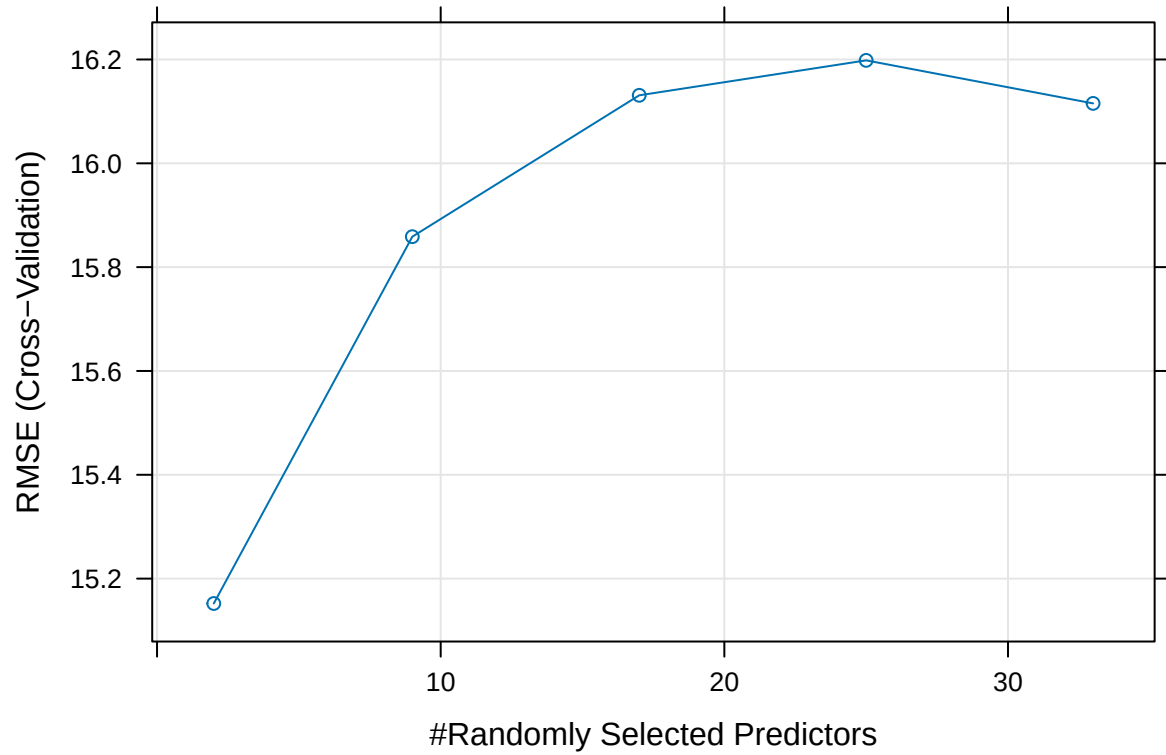
6.2.3 Limitations:

Random Forests sacrifice some interpretability compared to single trees. Additionally, since popularity is influenced by many external factors (e.g., marketing, social media), audio features alone can still only explain part of the variability.

6.2.4 Training:

To train the Random Forest model, we used the `caret` package, which streamlines model training and hyperparameter tuning. The `train()` function performs 5-fold cross-validation, where the dataset is split into five parts. Each part is used once as a validation set while the remaining four parts are used for training, rotating through all combinations. The `mtry` parameter — the number of variables randomly sampled at each tree split — was automatically tuned using `tuneLength = 5`, meaning 5 candidate values were tested.

The model was evaluated using Root Mean Squared Error (RMSE) across folds, and the best value of `mtry` was selected based on the lowest average RMSE. This helps balance bias and variance, improving generalization to unseen data.

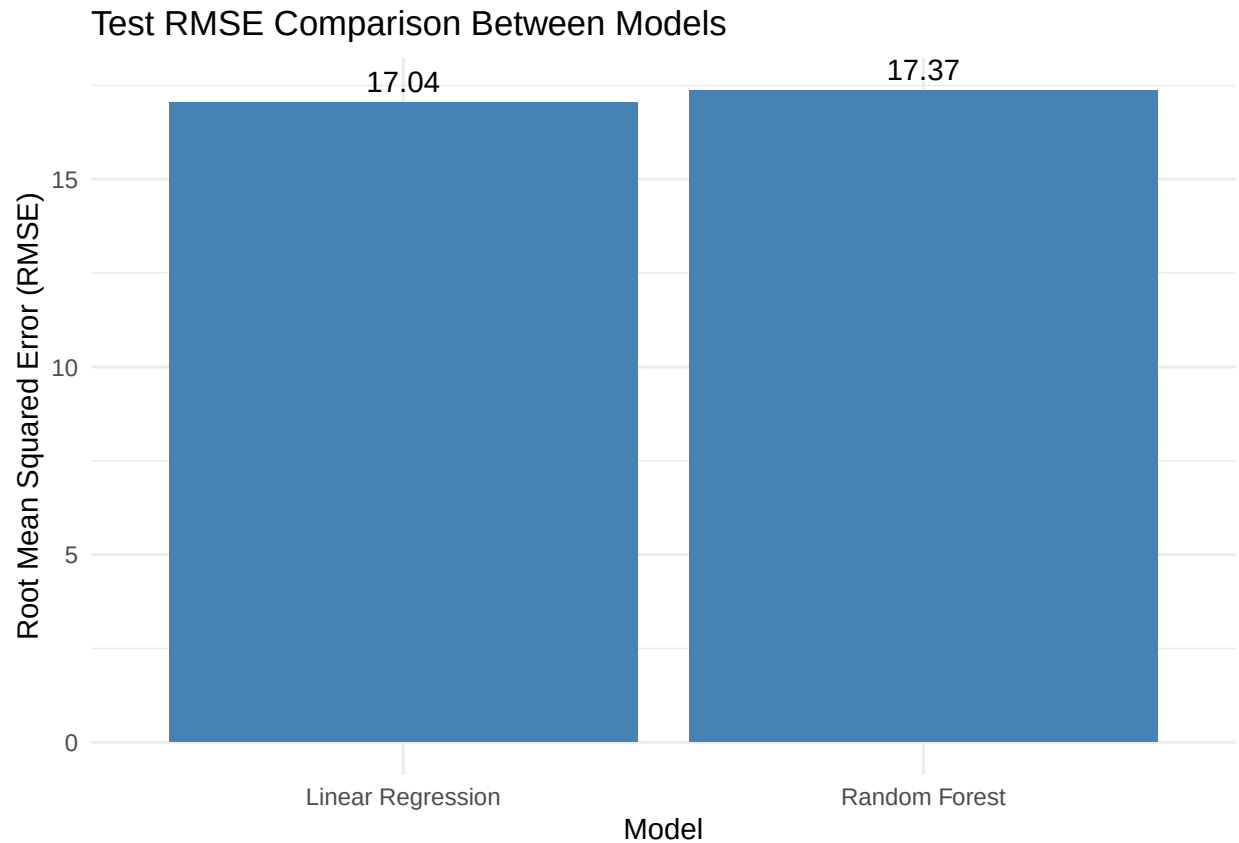


6.2.5 Final Random Forest Evaluation on Test Set:

The best tuned Random Forest model was evaluated on the held-out test set.

6.2.6 Final Comparison of Model Performance:

The test RMSEs for Linear Regression and Random Forest were compared to assess overall predictive performance.



6.2.7 Random Forest Interpretation

To improve prediction accuracy and capture complex, nonlinear relationships between audio features and track popularity, a Random Forest model was trained with hyperparameter tuning using cross-validation.

The tuning plot above shows how the test RMSE varied as the number of randomly selected predictors (`mtry`) changed. The lowest RMSE occurred when using only a small number of predictors, suggesting that selecting a focused subset of variables helped reduce overfitting and noise.

6.2.7.1 Final Model Performance

- **Test RMSE:** 17.37

This means that the model's average error in predicting popularity scores was approximately 17.37 units on the original scale — a slight improvement over the decision tree model but slightly worse than the regularized linear model (RMSE = 17.04).

6.2.7.2 Model Strengths

- **Captures nonlinear relationships** between predictors and popularity
- **Automatically models feature interactions**, such as combinations of genre and tempo or speechiness and loudness
- **Robust to multicollinearity** and irrelevant predictors

6.2.7.3 Model Limitations

- While flexible, the Random Forest model **sacrifices interpretability** compared to simpler models like linear regression or decision trees.
- The RMSE improvement is marginal, suggesting that increased model complexity does not drastically improve prediction — possibly due to limits in the explanatory power of audio features alone.
- Genre remains a dominant driver of predictions, as seen in variable importance rankings (not shown here), while most continuous features contribute modestly.

6.2.7.4 Takeaway The Random Forest model offered a flexible, ensemble-based approach to predicting Spotify track popularity. Although it did not outperform linear regression in this dataset, it validated that **interactions and nonlinear effects exist**, especially when accounting for categorical genres alongside audio characteristics.

7 Conclusion

This project explored the question: **What audio features are most predictive of a song’s popularity on Spotify?** By combining an open-source Kaggle dataset of Spotify tracks with real-time popularity data retrieved via the Spotify Web API, a diverse set of song-level features — including acoustic attributes and genre — were analyzed using multiple modeling approaches.

7.0.1 Summary of Findings:

- A **multiple linear regression model** identified **speechiness** as a significant negative predictor of popularity and highlighted **genre** as a strong determinant. Several genres — including *Children’s Music*, *Classical*, *Opera*, and *Reggaeton* — were significantly associated with lower popularity scores. Surprisingly, this linear model achieved the **lowest test RMSE of 17.04**, outperforming more complex models.
- A **regression tree** provided interpretable insights into how **genre**, **speechiness**, **energy**, and **liveness** interact to segment songs into low and high popularity groups. This supported earlier regression findings and emphasized that tracks with lower speech content and high energy generally perform better.
- A **random forest model**, tuned using cross-validation, achieved a slightly higher test RMSE of **17.37**. While it did not outperform linear regression, it confirmed the importance of **genre and energy**, and demonstrated that nonlinear methods can provide robustness and additional feature interaction insights.

7.0.2 Key Takeaway:

Across all models, **genre emerged as the most influential factor** shaping track popularity, often more so than audio features like danceability or valence. While some attributes such as **tempo** and **speechiness** showed consistent effects, their impact was secondary to genre.

This suggests that **musical success on Spotify is driven not just by sound properties**, but by broader contextual factors — including genre conventions, listener expectations, and possibly external dynamics like artist reputation or playlist inclusion — which were outside the scope of this analysis.

7.0.3 Future Directions:

- Incorporating **additional metadata** such as release year, playlist placements, or follower counts could improve predictive performance.
- Modeling with **XGBoost** or **neural networks** could help capture deeper nonlinearities and hidden patterns.
- Further analysis on **interactions between genre and features** (e.g., speechiness in rap vs. classical) could provide richer, genre-specific insights.

This analysis provides a strong foundation for understanding the complex drivers of music popularity and offers valuable direction for artists, producers, and data scientists alike.

8 Bibliography

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